

Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-03.jpg":

$$L = \lim_{n \rightarrow \infty} \left(n! \prod_{k=1}^n (\tan^{-1}(k+1) - \tan^{-1}(k)) \right)$$

Detailed Solution

To evaluate the limit L , we will analyze the behavior of the terms within the product and use the Squeeze Theorem to determine the final value.

Step 1: Bound the terms of the product

We first analyze the difference $\tan^{-1}(k+1) - \tan^{-1}(k)$.

By applying the Mean Value Theorem to the function $f(x) = \tan^{-1}(x)$ on the interval $[k, k+1]$, we know there exists some $c \in (k, k+1)$ such that:

$$\tan^{-1}(k+1) - \tan^{-1}(k) = f'(c) \cdot ((k+1) - k) = \frac{1}{1+c^2} \cdot 1$$

Since c is in the interval $(k, k+1)$, we have $c > k \geq 1$. This implies that $c^2 > k^2$, and consequently, $1+c^2 > k^2$.

Because $1+c^2 > k^2 > 0$, taking the reciprocal reverses the inequality:

$$0 < \frac{1}{1+c^2} < \frac{1}{k^2}$$

Substituting this back, we obtain a strict bound for the difference:

$$0 < \tan^{-1}(k+1) - \tan^{-1}(k) < \frac{1}{k^2}$$

Step 2: Bound the entire product

Since all terms in our product are strictly positive, we can safely multiply these inequalities from $k=1$ to n :

$$0 < \prod_{k=1}^n (\tan^{-1}(k+1) - \tan^{-1}(k)) < \prod_{k=1}^n \frac{1}{k^2}$$

We can simplify the upper bound by evaluating the product:

$$\prod_{k=1}^n \frac{1}{k^2} = \frac{1}{1^2 \cdot 2^2 \cdot 3^2 \cdots n^2} = \frac{1}{(n!)^2}$$

Thus, the finite product is bounded by 0 and $\frac{1}{(n!)^2}$.

Step 3: Apply the Squeeze Theorem

Now we multiply our inequality by $n!$ (which is strictly positive for all $n \geq 1$) to construct the full expression found inside the limit:

$$0 < n! \prod_{k=1}^n (\tan^{-1}(k+1) - \tan^{-1}(k)) < n! \cdot \frac{1}{(n!)^2}$$

Simplifying the upper bound yields:

$$0 < n! \prod_{k=1}^n (\tan^{-1}(k+1) - \tan^{-1}(k)) < \frac{1}{n!}$$

Finally, we take the limit as $n \rightarrow \infty$ across the inequality. The limit of the lower bound is trivially 0:

$$\lim_{n \rightarrow \infty} 0 = 0$$

The limit of the upper bound is also 0 because the factorial function grows without bound:

$$\lim_{n \rightarrow \infty} \frac{1}{n!} = 0$$

Since both the lower and upper bounds converge to 0, by the Squeeze Theorem, the limit of the expression between them must also be exactly 0.

Final Answer:

0